1. What is the difference between var, let and const? Explain scope, redeclaration and hoisting with a small example for each?

Var ->

Var is used local or global scope.

In this redeclaration possible.

Hoisting is done on it.

Eg.

Console.log(a,b); -> not showing error … it show undefined because of hoisting.

Var a = 10;

Console.log(a);-> 10 print

Var a = 20;

Console.log(a);-> 20 print – old value was destroyed by JS Engine. Safely created new value of a

Let ->

Let is used block scope.

In this redeclaration not possible.

Hoisting is done on it.

Eg.

{

Console.log(a); -> error initialise it than used

Int a =10

Console.log(a); -> print sussefully

}

Console.log(a); -> show error value a cannot reference.

Const ->

Similar to as let keyword but it hold the constant value

reassignment not possible.

Eg.

Const a =10;

a = 20; -> error cannot change constant value

JS Engine->

It is interpreter for a JavaScript. It jobs to convert the developer code into a machine language.

Hoisting ->

It is process of the execution the our Script by the JS

It have 3 level process.

1. Read Script -> In that JS Engine read the entirely script and finding the variable declaration and store it on browser.
2. Executed -> Than line by line code will be executes.
3. Output -> Then produces the output

2. What does typeof print for each of these: 4, "hello", true, null, undefined? Why is JavaScript called a dynamically typed language?

Solution:

typeof : It is a inbuild method or function to show your actual datatype of that particular function.

4 -> number

“hello” -> string

true -> Boolean

null -> object

Undefined -> undefined

Javascript is dynamically typed language because variable do not need explicit the type decalartion and values will be change different data type at runtime . JS Engine can determine it and executes.

3.Explain == versus ===. Then explain the difference between || and ??, and say which one you would use for a default value when 0 is a valid input.

Solution

== -> compare only the based on value
eg. 5 == “5” it can upcasting the “5” string to number 5 than compare return true

=== -> compare both based on value and type

Eg. 5 === “5” it can strictly check type also and value true

|| -> 0,””,false,null,undefined right left hand side which is used || operator return default value

?? -> null,undefined left hand side than return the default value other is valid 0,””,false

4.

A server sends you a student with an id, a name, two phone numbers and an address that has a street and a city. Write it as a JSON object. Show how you read the second phone number and the city. What does student?.address?.city do when address is missing?

```
Const studentData = {
        Id : 101,
        Name : “ritik”,
        Phone : [232432123,45532122],
        Address: {
                Street:“Devpur”,
                City:”Dhule”
        }
};
Console.log(studentData.phone[1]);
Console.log(studentData.address.city);
```

No, is not missing city it definitely show city it goes first from student -> address -> city

5. Given var arr = [10, 45, 32, 76, 45, 61, 78];, write one line each to get (a) the even numbers, (b) every number doubled, (c) the sum of all numbers.
var arr = [10, 45, 32, 76, 45, 61, 78];

```
arr.forEach((n)=> console.log(n));


let even = arr.filter((n) => n % 2 == 0);
let product = arr.map((n) => n*n);
let sum = arr.reduce((n)=>n+=n);
console.log(even,product,sum);
```

6.Explain what a callback function is. Rewrite function add(x, y) { return x + y; } as an arrow function with a single statement.

Callback function :-> Javascript allow to send a function definition as a argument to an other function this process known as callback function

```
let add = (a,b)=>{
  return a + b;
};


console.log("Sum is",add(10,20));
```

7. What is the output order of this code, and why? console.log("A"); setTimeout(() =>

console.log("B"), 1000); console.log("C");

A

C

B

It is by default synchronusing process. It can not block the execution. At once they can start execution. For this example A,C immediately print but B is have setTimeout method so it will hold execution 1s after that it print. So , A C B output

8. What is callback hell, and how do promises solve it? Name the three states of a

promise and the two methods used to handle it.

-> When they function is depending each other that time we call the function inside function so that make very complex to a function. That time arise a callback hell problem.
-> Using the promises you can make a seaprate function to implemented logic inside than simply return the promises.
-> there are three state of promises
1. pending -> by default pending

2. fulfill/resolve -> after successfully compliation

3. reject -> error occurs go to reject state

-> there are two metheod to use
1. .then() -> responsible for showing resolve state and handle it.

2. .catch() -> responsible for catching the error or reject state and handle it.

9. Rewrite sum(2, 3).then(r => console.log(r)).catch(e => console.log(e)) using async and

await with try and catch. Why must the function be marked async?

```
async function sum(a,b){
    return new Promise((resolve,reject)=>{
      resolve(a+b);
    });
  }

  async function handleSum() {
    try {
    let result =  await sum(2,5);
    console.log(result);
   } catch (error) {
    console.log(error);
   }

  }

  handleSum();
```

10. What is the DOM? Name the method used to find an element by its id, and explain

the difference between innerHTML and value.

-> DOM : DOM is document object module. It that we can change the styling and content dynamically using document predefined object in js.

Suppose we have p tage

Hi, I am Ritik 😊

I want to first access this tag than perform some operation so for that document is help us.

Document.getElementByID(“para”);//inside “” we can pass actual ,p tag id

Difference is

innerHTML -> give the inside tag connect present or we can added also inside in

eg, ,<h> etc

value -> value is actual value which giving by user inside input

eg, form

11. What does fetch(url) return, and why do we need await twice, once for fetch and

once for response.json()? What does response.status tell you?

-> fetch() is method that can help to you connect with the server for the featching response

-> it give 2 argument , fetch(url,method)

->URL-> Is your server url https api or url

->METHOD-> Is also menthod to along it to know fix what you what like retrive data to used GET, update or submit data to used it POST. It is like multiple define menthod GET,POST,DELETE ect

-> why 2 twice await is return promises of data it contains all meta information like satus code, url , etc for this to handle this data so we want to a await there, and rather the main data data will inside that contents in json(), type so again featching this from the content also it return a new promise also for this handle we need await and that json() data will become a actual.

-> the response.status is predefined code which trigger at specify error and also have specified meaning on it. Using this server and client communicated if status:200 that mean client get ok , and give data

Susses code start range of 100,200,300 and failture 400,500

12. What is a regular expression, and why is it better than many if statements for checking a password rule?

- ➔ It is regex , containing the sequence of character that can matching the existing input which is giving by the user and based on that we can achieve validation like a password should contain one uppercase, one lowercase, one special symbol and length should be 8 minimum
- ➔ Regex is better than if statement because we can added a multiple condition and also increases the complex code of line rather than used a regex